PR-2

Part-1

1) what is Inferential Statistics
→ Inferential Statistic ~~statistical method~~ k is a brach of Statistics that uses sample data to draw Conclusions, make predictions and infer characteristics about a larger population.

2) what is Hypothesis Testing and its Components?
→ Hypothesis testing is a Statistical method used to determine whether there is enough evidence to Support a claim about a population.

Components
- Null Hypothesis (H_0)
- Alternative Hypothesis (H_1)
- Significance Level (α)
- Test Statistic
- Critical value
- p-value
- Decision Rule

3) Explain Confidence Interval and Critical Value.
→ ~~Co value~~

Confidence Interval (CI):
A range of values that is likely to Contain the true population parameter with a Specified Confidence level (eg. 95%)

Critical value:
A threshold value used to determine whether the null hypothesis should be rejected

4.) Define p-value

→ P-value is the probability of obtaining results at least as extreme as the observed results, assuming the null hypothesis is true

5.) Differentiate Type I and Type II Errors

Error Type	Description
Type I Error	Rejecting a true null hypothesis.
Type II Error	Failing to reject a false null hypothesis

6.) Brief Description of Statistical Tests

-> Z-Test: Used to compare means when the sample size is large and population variance is known.

T-Test: Used to compare means when the sample size is small and population variance is unknown.

Chi-Square Test: Used to determine whether there is an association between categorical variables

ANOVA (Analysis of Variance): Used to compare means across three or more groups

7) what is Covariance?
→ Covariance measure the direction of the relationship between two variable

8) What is Correlation?
→ Correlation measures both the strength and direction of the relationships between two variable. Its value ranges from -1 to +1

Part-B

1) Hypothesis 1: Smoking vs Diabetes

H_0: Smoking has no effect on diabetes
H_1: Smoking affects diabetes

Result: p-value: 0.0035
Conclusion: Since p-value < 0.05, reject H_0.
Smoking is significantly associates with diabetes.

2) Hypothesis 2: Exercise vs BMI
H_0: Exercise frequency has no effect on BMI
H_1: Exercise frequency affects BMI
Result: p-value: 0.00012

Conclusion, since p-value < 0.01, reject H_0

Exercise frequency significantly affects BMI

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3.) Confidence Intervals
· Age (95% CI): 40.8 - 44.4 years
· weight (95% CI): 69.4 - 73.2 Kg

Interpretation:- The true population means are likely to lie within these ranges

4.) T-Test (Blood Pressure by Gender)
Result: p-value = 0.007
Conclusion: Reject H_0. Blood pressure differs significantly between male and females

5.) chi-Square Test
Result: X^2 11.28, p-value = 0.0035
Conclusion: Smoking status and diabetes are significantly related

6.) ANOVA Test
Result: F = 5.92, p-value = 0.0012
Conclusion: BMI differs significantly across exercise groups

7.) Covariance
Age vs BMI = 18.72
Interpretation: Age and BMI tend to increase together

8.) Correlation
Age vs BMI = 0.41
Interpretation: There is a moderate positive relationship between Age and BMI